

Kyle Yu

2770 Canterbury Rd. San Marino, CA 91108

(626) 688-4331 | kyle.yu45@gmail.com | [linkedin.com/in/kylecyu](https://www.linkedin.com/in/kylecyu) | github.com/kyolebu | kylecyu.com

EDUCATION

Boston University Faculty of Computing & Data Sciences; Boston, MA

Jan 2023 – May 2026

Bachelor of Science in Data Science

Minor in Computer Science

Cumulative GPA: 3.59

Societies: Club Volleyball President & Captain, 180 Degrees Consulting Director of Finance, Kappa Theta Pi

WORK EXPERIENCE

Red Hat; Boston, MA

May 2025 – Present

Software Engineering Intern – Emerging Technologies

- Engineering Triton GPU kernels to accelerate inference and training pipelines for large language models
- Exploring vLLM's architecture and Triton-based acceleration strategies to contribute performance improvements for LLM inference on NVIDIA and AMD GPUs, with a focus on distributed caching

Shuum; Palo Alto, CA

Sep 2024 – Dec 2024

Software Engineering Intern – AI/ML

- Collaborated with engineering team to design and implement a recommendation engine powered by vector similarity search, leveraging Pinecone for a vector database and LangChain for building ranking workflows
- Created a data model to streamline structured data flow, enabling improved recommendation relevance

Human-to-Everything Lab; Boston, MA

Nov 2024 – Dec 2024

Undergraduate Researcher

- Building and running simulations in CARLA to replicate real-world driving scenarios on Boston road systems to train machine learning models for autonomous vehicle perception, decision-making, and control

Headstarter AI; Long Island City, NY

Jul 2024 – Sep 2024

Software Engineering Fellow

- Developed RAG system with web scraped data and Pinecone to improve quality of generated text by LLMs
- Finalist in Retell AI Hackathon for building a real-time conversational voice agent with Whisper speech-to-text, WebSockets for low-latency communication, Llama 3.1 as the LLM, and Tortoise text-to-speech

Milliman; Orange County, CA

Nov 2022 – Jun 2023

Actuarial Data Processing Intern

- Developed 750+ dynamic, data-driven rate tables to calculate premiums, financial costs of risk, and uncertainty for auto and homeowner insurance, establishing the foundation for future actuarial analysis
- Reviewed 500+ rate tables for technical accuracy to prepare work for eventual coded rate work

PROJECTS

Basketball CV Analysis System – Python, OpenCV, YOLO, KNN

Jun 2024 – Aug 2024

- Uses machine learning, computer vision, and deep learning to analyze player performance and game dynamics
- Trained a custom object detector in YOLO using annotated detection data from Roboflow and implemented it in Python to achieve enhanced accuracy in detecting players, referees, and basketballs in gameplay footage

COVID-19 Policy Evaluation – Python, Azure, Power BI, SQL/DAX

Mar 2024 – May 2024

- Integrated data from three sources into a centralized data lake using Azure Data Factory for the data pipeline
- Transformed the data into Fact and Dimension tables in an operational data store, externalized them in Azure Synapse for processing, and built a star schema in Power BI to support accurate and structured analysis

SKILLS AND INTERESTS

Coding: Python, Java, SQL, Javascript, Rust, HTML, CSS

Libraries/Frameworks: Triton, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, SciPy, TensorFlow, RAG

Tools: Microsoft Azure, AWS, Power BI, Git, Microsoft Excel, Pinecone, LangChain, APIs, Kubernetes, Docker

Interests: Sports, Philosophy Books, Skiing, Fishing, Traveling, Stocks, Crypto